

Adaptive Suppression and Enhancement in Auditory Search

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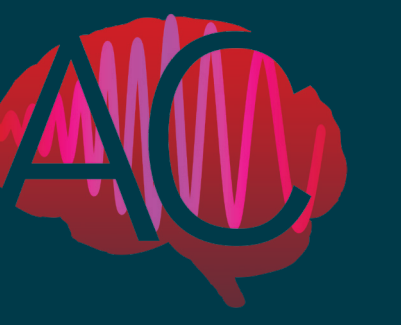
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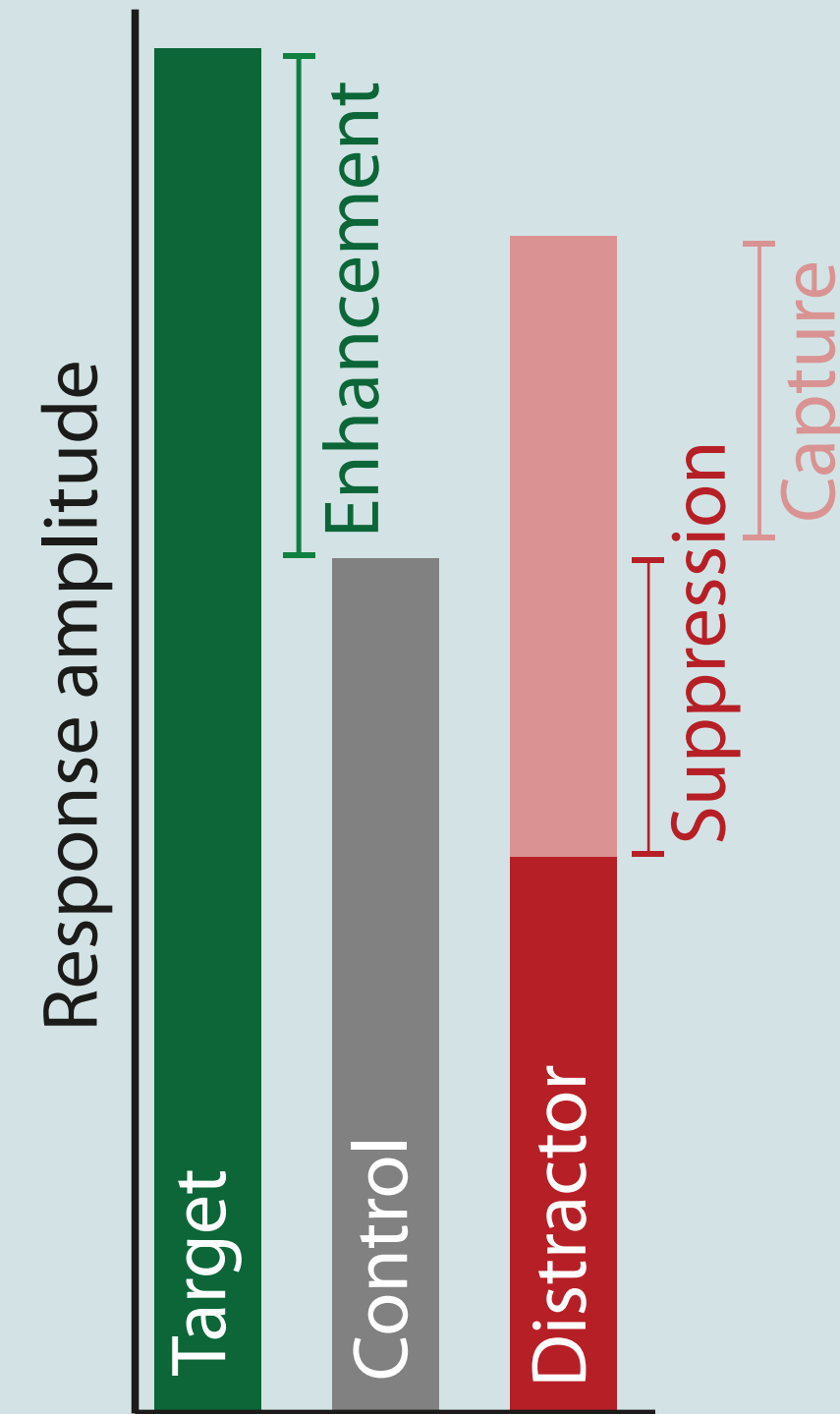
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Background

Human environments consist of relevant targets and irrelevant distractors. In auditory attention research, the understanding of capture and suppression is premature, partly because target and distractor effects are often confounded¹.

Research goal: We introduce a baseline to directly compare neural and behavioral responses between control versus target and distractor sounds, inferring mechanisms of target **enhancement**, distractor **suppression** and **capture**².



Design and Target Towardness

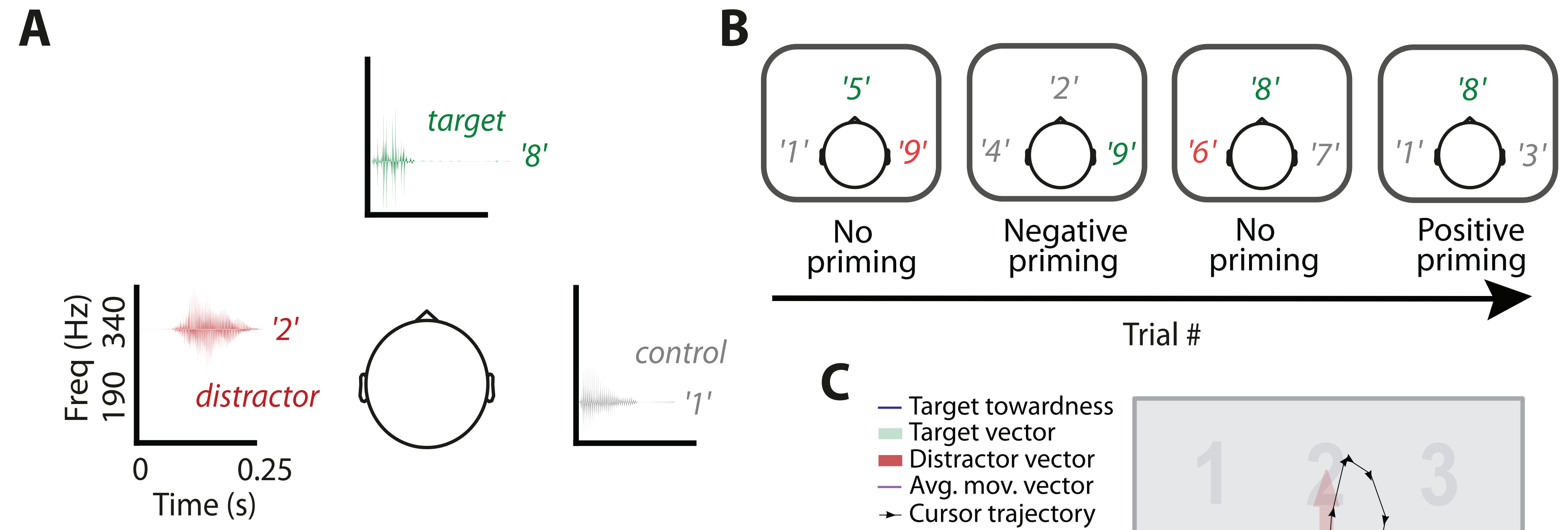


Figure 1: Experiment design and cursor trajectory analysis. **A)** Speaker arrangement and categorical sound characteristics. Three simultaneous one-syllable digits are presented from spatial positions -90° , 0° , and $+90^\circ$ relative to the participants' head. **B)** Target repeats (positive priming) and distractor-target switches (negative priming) within a sequence of four consecutive trials³. **C)** Trial-wise cursor trajectories were averaged (purple arrow). Target cursor towardness (blue arrow) was calculated as the scalar product of the average movement vector and the target vector (green arrow; 0.6 dva). The figure shows a trial in which the initial movement direction aimed towards the salient distractor (red arrow). Button box size was 2 dva.

Results

Target Enhancement

Distractor Suppression

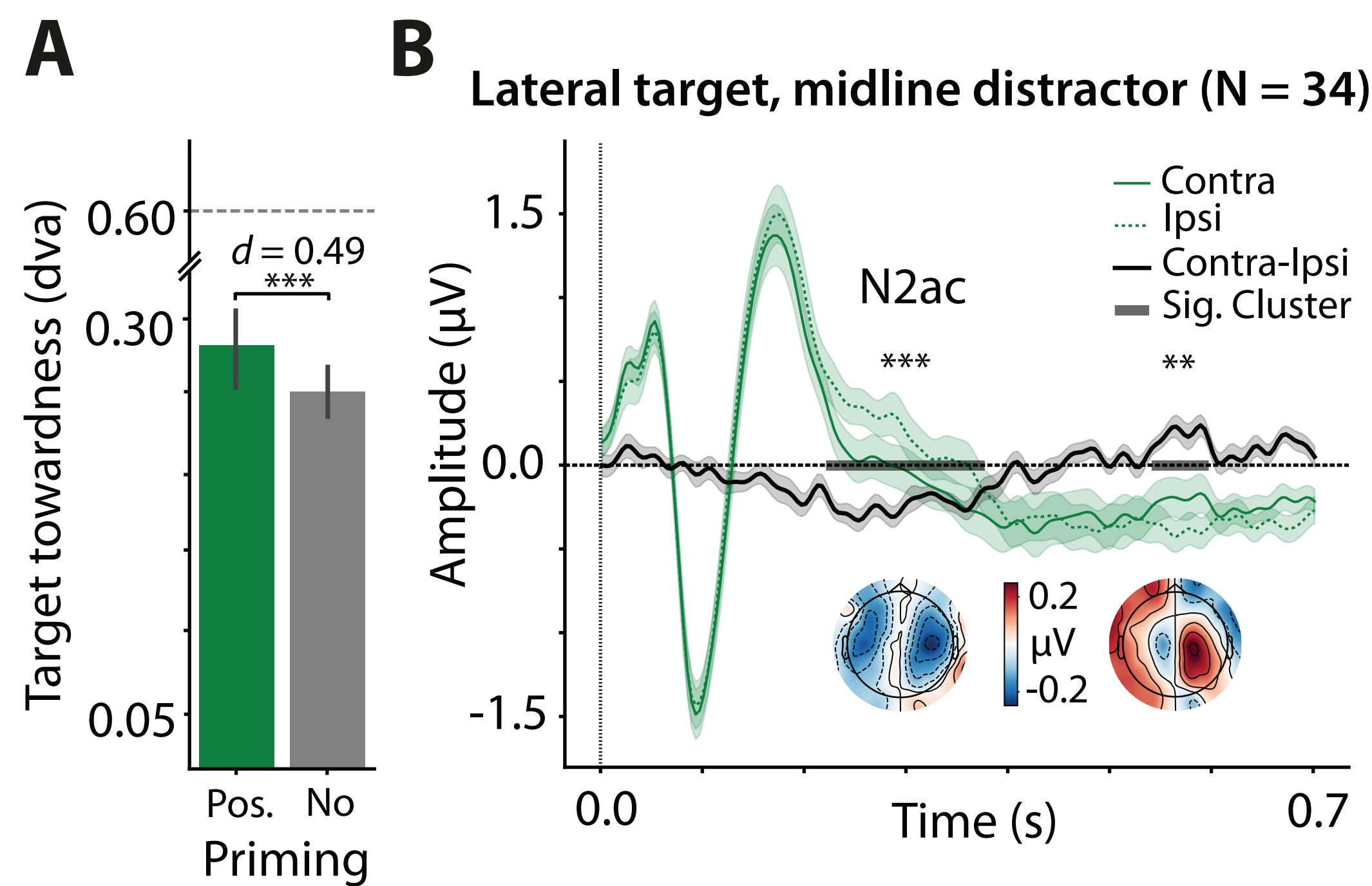


Figure 2: Signatures of target enhancement. **A)** Behaviorally, target enhancement was expressed as heightened target towardness in target-repeat trials ($p < 0.001$). Dashed grey line represents optimal performance. **B)** In the EEG, target enhancement was observed by N2ac emergence in lateralized target trials at electrode pair C3/4 ($p = 0.001$). A later component was observed ($p = 0.006$), possibly reflecting spatial reorienting^{4,5}. Inset topographies show difference wave distribution for significant cluster. Shaded areas represent ± 1 s.e.m.

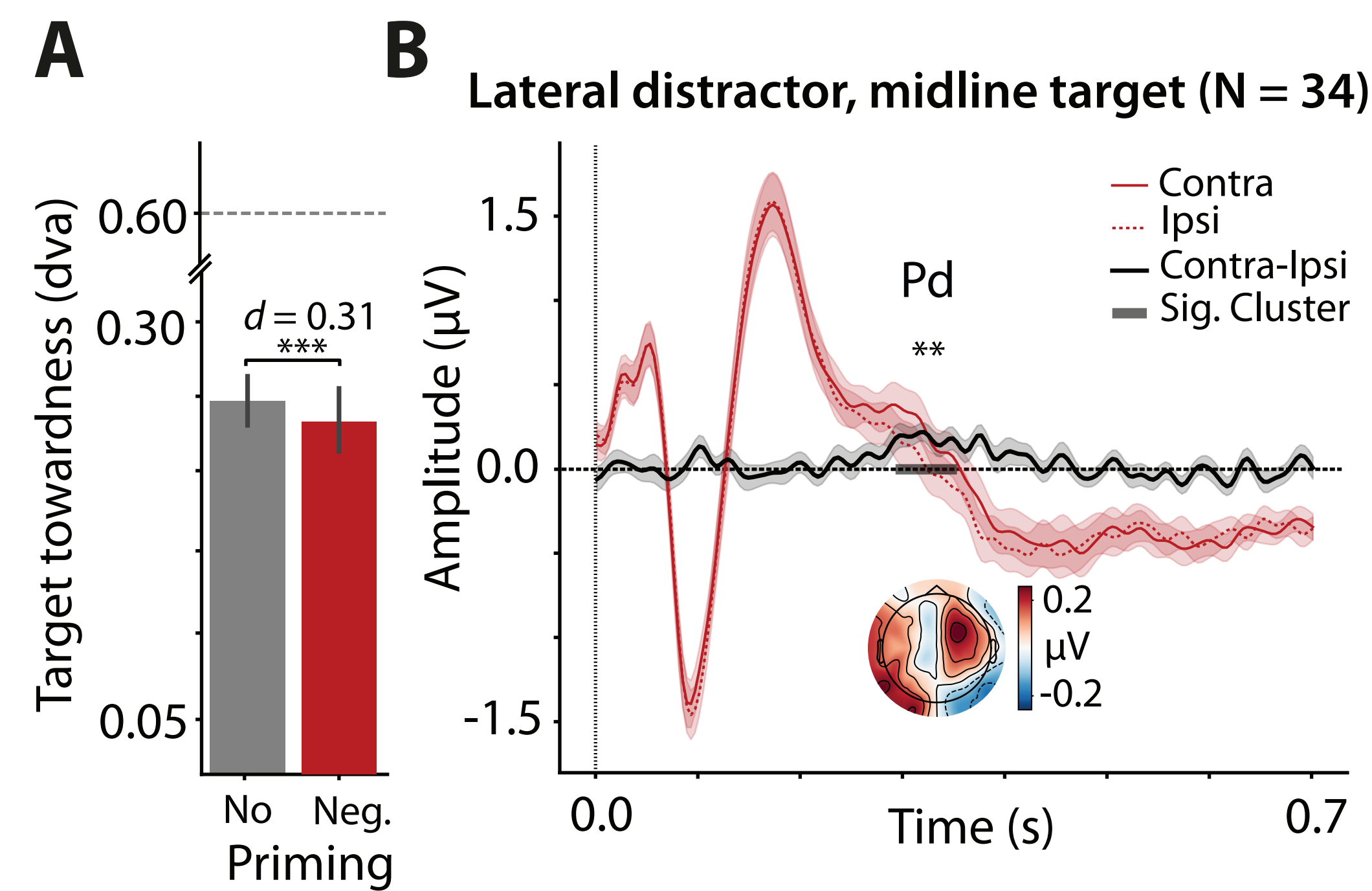
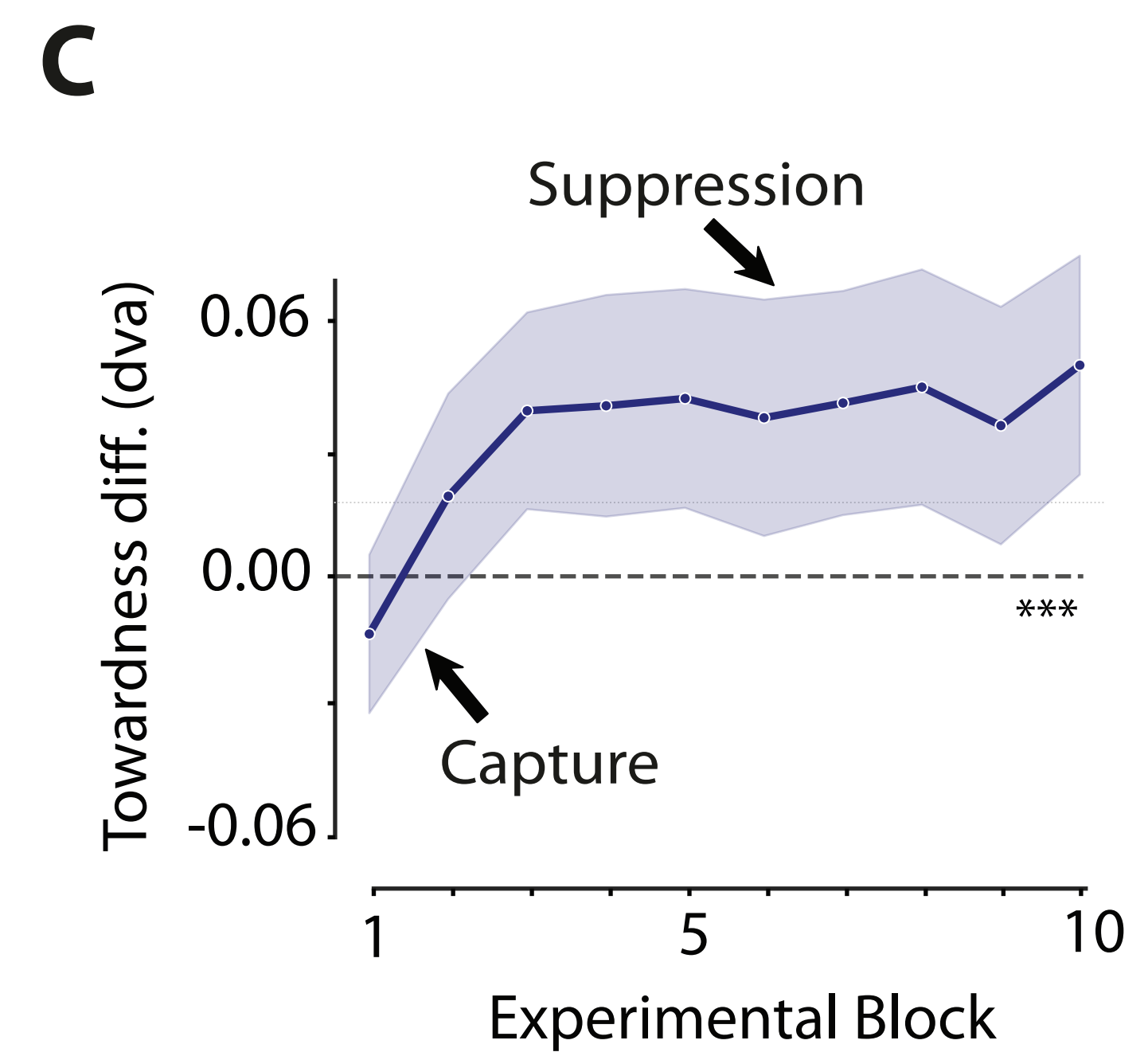


Figure 3: Signatures of distractor suppression. **A)** Behaviorally, distractor suppression was expressed as decreased target towardness in distractor-target-switch trials ($p < 0.001$). Dashed grey line represents optimal performance. **B)** In the EEG, distractor suppression was observed by Pd emergence in lateralized distractor trials at electrode pair C3/4 ($p = 0.004$). Inset topographies show difference wave distribution for significant cluster. **C)** Over the course of the experiment, target towardness relative to distractor towardness increased, indicating a gradual transition from attentional capture to more successful distractor suppression⁶ (Page's trend test: $L = 10758$, $p < 0.001$). Shaded areas in **B)** and **C)** represent ± 1 s.e.m.



Neural Activity Relates to Target Towardness

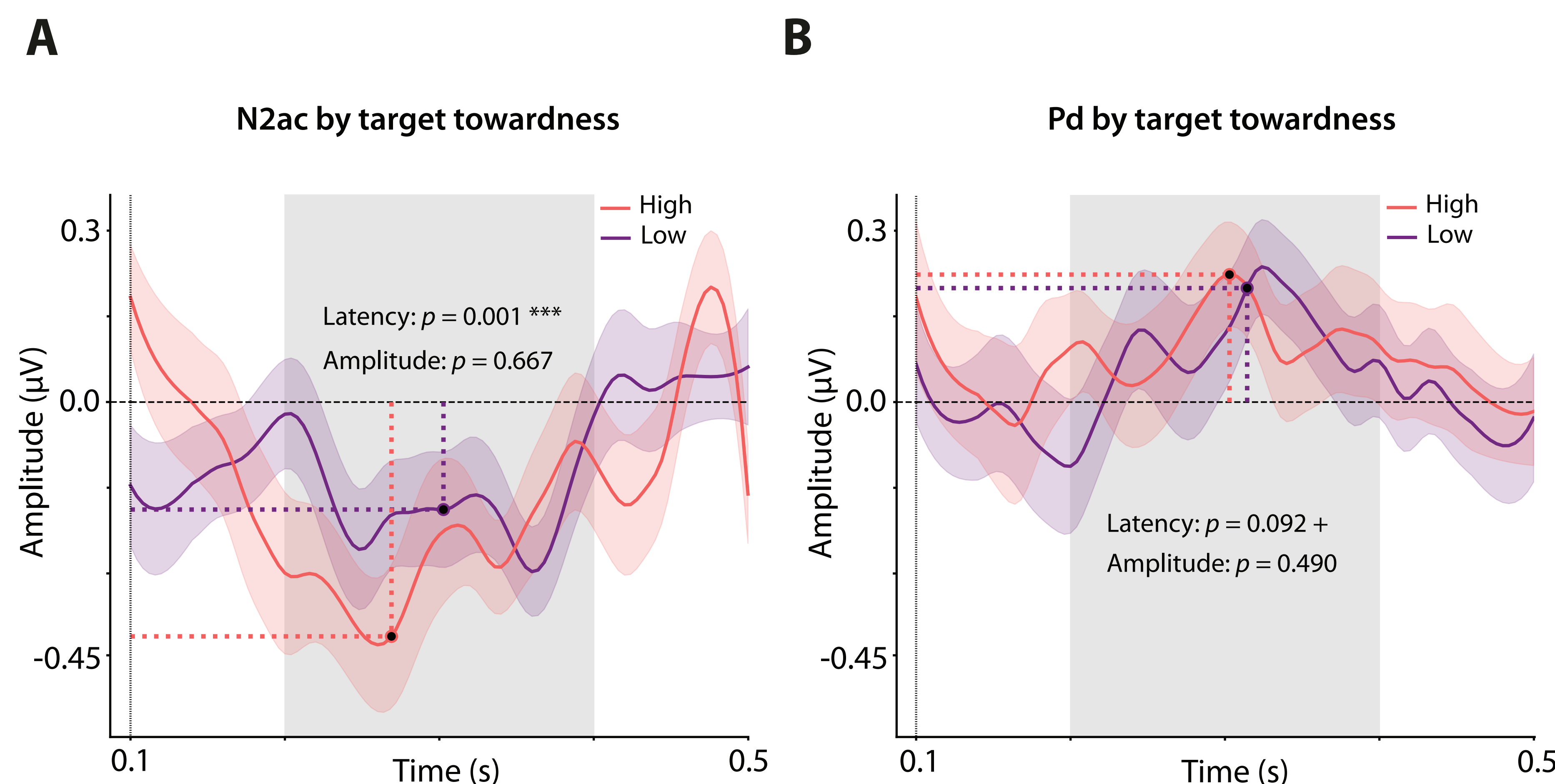


Figure 4: Target towardness relates to ERP components. Note that both panels visualize identical analysis on lateralized target (N2ac) and distractor (Pd) trials, respectively. Single-trial component latencies and amplitudes were retrieved between 0.2 - 0.4 s post-stimulus onset as the absolute 50% cumulated voltage. P values were estimated by a linear mixed-effect model and equated target_towardness ~ (1|Subject ID) + trial number + amplitude + latency. The median-split target towardness serves solely illustration purpose. A Savitzky-Golay filter (order: 3; frame length: 21) was applied to the waveforms before visualization to improve the visibility of effects. **A)** Small N2ac latencies relate to greater target towardness compared to large N2ac latencies. The same trend can be observed for lateralized distractor trials in **B)**. Shaded areas in both panels represent ± 1 s.e.m.

Conclusion

1. Our design offers a novel approach to draw the established visual search paradigm into the auditory modality, delineating mechanisms of target enhancement from distractor capture and suppression. This design enables the study of reactive auditory attention, which has been widely neglected in the past.
2. We establish a novel metric into auditory search, namely target towardness. This measure effectively captures the above mentioned dynamics of selective attention, negating constraints of conventional metrics.
3. Functionally, participants exhibit an auditory distractor positivity (Pd) to suppress a salient distractor. This suppression signature tends to relate to target towardness.

References

- [1] Wöstmann et al. (2022), Progress in Neurobiology.
- [2] Orf et al. (2023), iScience.
- [3] Tipper (1985), The Quarterly Journal of Experimental Psychology.
- [4] Gamble & Luck (2011), Psychophysiology.
- [5] Gamble & Woldorff (2015), Cerebral Cortex.
- [6] Gaspelin & Luck (2017), Journal of Experimental Psychology.

